

GNU nano 8.7.1

task1.c

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    else if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());

        printf("Executing command: %s\n", command);
        execlp(command, command, NULL);

        printf("Command execution failed!\n");
    }

    else {
        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }
}
```

```

siddu@DESKTOP-SCRM0SP:~$ uname -a
Linux DESKTOP-SCRM0SP 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
siddu@DESKTOP-SCRM0SP:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      12
On-line CPU(s) list:        0-11
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i5-13420H
CPU family:                  6
Model:                       186
Thread(s) per core:         2
Core(s) per socket:         6
Socket(s):                   1
Stepping:                    2
BogoMIPS:                    5222.42
Flags:                        fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc deadline timer aes xsave avx f16c rdr
                             and hypervisor tshf tsx bmi bmi1 avx2 smep bmi2
                             bts lbpb stibp lbrs enhanced_tor_000efetch sopl
                             d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bti
                             sha_ni xsaveopt xsavecsgsbs bti oes1a nopt0nwb
                             vnni umip waitpkg gfni clx xgetbv1 vsauidq avx_vnni
                             bb_cll noindir64 bsr md_clear gerlllize lb lbt flu
Virtualization features:
Virtualization:              VT-x
Hypervisor vendor:           Microsoft
Virtualization type:         full
Caches (sum of all):
L1d:                          288 KiB (6 instances)
L1i:                          192 KiB (6 instances)
L2:                            7.5 MiB (6 instances)
L3:                            12 MiB (1 instance)
NUMA:
NUMA node(s):                1
NUMA node0 CPU(s):           0-11

```

```
siddu@DESKTOP-SCRM0SP: X  Ubuntu  X  Ubuntu  X  Ubuntu  X  +  -  □  X

Vulnerabilities:
Gather data sampling:    Not affected
Ghostwrite:              Not affected
Indirect target selection: Not affected
Itlb multihit:          Not affected
L1tf:                   Not affected
Mds:                    Not affected
Meltdown:               Not affected
Mmio stale data:        Not affected
Old microcode:           Not affected
Old file data sampling:  Mitigation: Clear Register File
Retbleed:                Mitigation: Enhanced IBRS
Spec rstack overflow:    Not affected
Spec store bypass:       Mitigation: Speculative Store Bypass disabled via prctl

Spectre v1:              Mitigation: usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:              Mitigation: Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S

Srbds:                  Not affected
Tsx:                    Not affected
Tsx async abort:         Not affected
Ldscape:                 Not affected

siddu@DESKTOP-SCRM0SP: ~$ ps
  PID TTY          TIME CMD
 14448 pts/6      00:00:00 bash
 14491 pts/6      00:00:00 ps
```

```
siddu@DESKTOP-SCRM0SP:~$ top
top - 15:07:15 up 1:19, 1 user, load average: 0.00, 0.02, 0.07
Tasks: 43 total, 1 running, 42 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.3 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 7776.6 total, 7061.2 free, 653.0 used, 229.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7123.7 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
  113 root        20   0   35376   12302   9232 S   0.3   0.2   0:03.33 systemd-udevd
14503 shreyaj    20   0    8320    6148   3952 R   0.3   0.1   0:00.06 top
    1 root        20   0   24480   15356   11312 S   0.0   0.2   0:03.63 systemd
    2 root        20   0    3180    2212   1272 S   0.0   0.0   0:00.02 init-systemd(ub
    7 root        20   0    3228    2160   2008 S   0.0   0.0   0:00.63 init
   72 root        19  -1   58408   17052   15784 S   0.0   0.2   0:00.47 systemd-journal
  107 systemd+   20   0   22540   14580   11956 S   0.0   0.2   0:00.15 systemd-resolve
  135 root        20   0    2888    1948    1820 S   0.0   0.0   0:00.12 chronyd-starter
  136 root        20   0    4172    3172   2924 S   0.0   0.0   0:00.02 cron
  137 message+  20   0    9040    8596   4780 S   0.0   0.1   0:00.73 dbus-daemon
  143 root        20   0   44092   29724   14576 S   0.0   0.4   0:00.85 networkd-dispat
  175 root        20   0   18644    9512   8184 S   0.0   0.1   0:00.49 systemd-logind
  265 _chrony     20   0   20420   6365   3584 S   0.0   0.1   0:00.17 chronyd
  277 _chrony     20   0   12492    2464   1824 S   0.0   0.0   0:00.01 chronyd
  311 root        20   0   123456   32700   1092 S   0.0   0.4   0:00.52 unattended-upgr
  313 root        20   0    2492    2704   18092 S   0.0   0.1   0:00.01 apetty
  408 shreyaj    20   0   22468   12744   8008 S   0.0   0.2   0:00.01 logind
  412 shreyaj    20   0   22068    5464   3016 S   0.0   0.1   0:00.01 (sd-pam)
  436 shreyaj    20   0    6134    5424   3864 S   0.0   0.1   0:00.03 bash
  773 root        20   0    3180    1260   1188 S   0.0   0.0   0:00.06 SessionLeader
  778 shreyaj    20   0    2200    1260   1188 S   0.0   0.0   0:00.09 Relay(778)
1258 root        20   0    6284    5544   3972 S   0.0   0.1   0:00.17 bash
1259 root        20   0    3184    1124   1908 S   0.0   0.0   0:00.00 SessionLeader
1264 shreyaj    20   0    6208    1608   1188 S   0.0   0.0   0:00.01 Relay(1264)
1752 sbot        20   0    6268    5604   3936 S   0.0   0.1   0:00.05 bash
1753 shreyaj    20   0    3200    1264   1188 S   0.0   0.0   0:00.08 SessionLeader
14168 root        20   0    6268    1800   1188 S   0.0   0.0   0:00.01 Relay(1758)
```

```
siddu@DESKTOP-5CRM05P: ~$ mkdir OS_LAB
mkdir: cannot create directory 'OS_LAB': File exists
siddu@DESKTOP-5CRM05P: ~$ cd OS_LAB
siddu@DESKTOP-5CRM05P: ~/OS_LAB$ nano task1.c
siddu@DESKTOP-5CRM05P: ~/OS_LAB$ ^C
siddu@DESKTOP-5CRM05P: ~/OS_LAB$ gcc task1.c -o task1
siddu@DESKTOP-5CRM05P: ~/OS_LAB$ sudo apt update
[sudo] password for siddu:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
25 packages can be upgraded. Run 'apt list --upgradable' to see them.
build-essential is already the newest version (12.12ubuntu2.26.04.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 25
siddu@DESKTOP-5CRM05P: ~/OS_LAB$ ./task1
Enter a Linux command: ls
Parent Process
Parent PID : 14386
Child PID  : 14387
Child Process
Child PID command: ls
task1 task1.c
Child process completed.
siddu@DESKTOP-5CRM05P: ~/OS_LAB$ pwd
/home/siddu/OS_LAB
siddu@DESKTOP-5CRM05P: ~/OS_LAB$
```